Part C: Electron Configuration

12. How many electrons can each level hold? 1st = _____ 2nd = _____ 3rd = _____

13. What term is used for the electrons in the outermost shell or energy level? _____

14. Scientists use two types of diagrams to show the electron configuration for atoms. Follow your teacher's directions to complete the diagrams.

Sulfur

Atomic # = 16

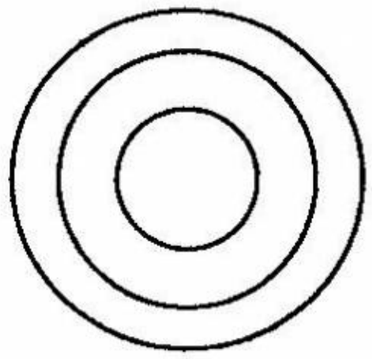
Atomic Mass = 32

Protons = ____

Neutrons = ____

Electron = ____

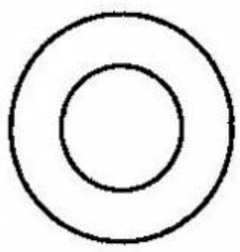
Bohr Diagram
Shows <u>all</u> electrons



Lewis Structure
Shows <u>valence</u> electrons

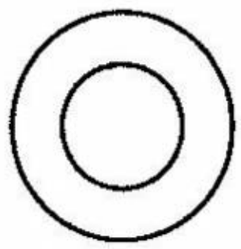
S

15. Calculate the missing information and then draw the Bohr Diagram and Lewis Structure for each element.



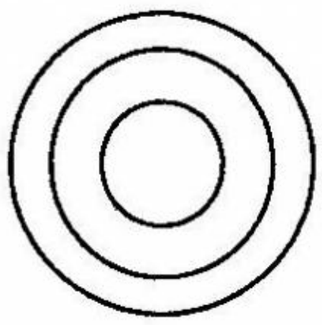
Atomic # = 3
Mass # = 7
of P = ____
of N = ____
of E = ____

Li



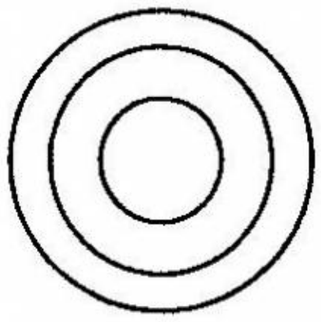
Atomic # = 10
Mass # = 20
of P = ____
of N = ____
of E = ____

Ne



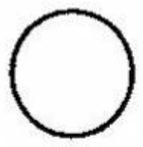
Atomic # = 12
Mass # = 24
of P = ____
of N = ____
of E = ____

Mg



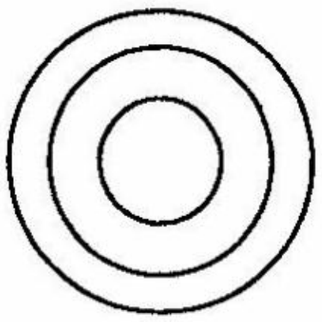
Atomic # = 17
Mass # = 35
of P = ____
of N = ____
of E = ____

Cl



Atomic # = 2
Mass # = 4
of P = ____
of N = ____
of E = ____

He



Atomic # = 14
Mass # = 28
of P = ____
of N = ____
of E = ____

Si

16. Answer the questions below based on the elements in question #15.

(1) Which elements had a filled outermost shell? _____ _____

(2) Which element would be most likely to lose electrons in a chemical bond? _____

(3) Which element would be most likely to gain electrons in a chemical bond? _____

(4) Which elements are not likely to bond with other elements? _____ _____ Why? _____